

Predictors as Markets

Coherence without convexity, frictions as regularizers, and the market as biconjugate

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Abstract

Many learning procedures are cost-function markets by identity rather than analogy: mirror descent trades against a convex-cost maker, Bayesian model averaging is a market of Kelly bettors with wealths as posterior weights, least squares is the precision-weighted merge of one quadratic maker per observation, and a proximal step is the optimal response to a maker charging the penalty. This note asks the converse, which predictors are markets, and finds that the apparent obstruction, non-convexity, does not obstruct coherence: charges telescope for any cost, and the exact no-arbitrage condition is that chord slopes stay in the convex hull of payoffs. Rational flow trades the biconjugate, landing on the contact set with profit equal to the envelope profit plus the starting gap, so non-convexity costs expressiveness, unquotable states that read as holes in the book and multimodal quotes, rather than soundness. Frictions price the remaining failure: a proportional fee of at least the chord excursion restores no-arbitrage, the mechanism-level analogue of no-arbitrage under small transaction costs, and the fee-spread lemma survives non-convexity with the envelope in place of the cost; a deep quadratic co-quoter drives the merged venue's incoherence to zero like one over its depth via the Moreau envelope. A predictor is a market when its arbitrage depth is finite relative to affordable friction; convexity is the zero-friction case.

1 The correspondences, and the question

A cost-function market maker posts a potential C over inventory and charges $C(q + s) - C(q)$ for a fill s ; prices are gradients and, when C is convex, the standard theory applies (Hanson 2003; Abernethy et al. 2013). A run of published identities makes many learning procedures markets in this literal sense. Trading against such a maker is follow-the-regularized-leader with the cost as conjugate regularizer (Chen and Vaughan 2010; Abernethy et al. 2013), with trader wealth as the learning rate (Frongillo et al. 2012). A market of Kelly bettors performs Bayesian model averaging with wealths as posterior weights (Beygelzimer et al. 2012), and equilibrium prices of utility-maximizing agents implement mixtures and products of experts (Pennock and Wellman 1997; Storkey 2011; Storkey et al. 2012; Barbu and Lay 2012). Ordinary least squares is a market of data points: each observation a quadratic maker quoting its value with capital equal to its precision, the estimate their merge, since merging makers is the infimal convolution of their costs and liquidity adds (Bhaskara et al. 2023). A proximal step is a trade against a fee-bearing maker: the proximal operator of $f|\cdot|$ is the soft-threshold, which is the optimal response to a proportional fee, and the prox of any convex g is the response to a maker charging g . The two canonical penalties are then the two market primitives: ridge is a zero-quoting participant with capital λ , lasso is a fee of λ , and the theorem that regularization is robustness to data

perturbation (El Ghaoui and Lebrete 1997; Xu et al. 2009) acquires a market reading in which the adversary’s budget is priced rather than assumed.

The question is the converse. Which predictors are markets, and what exactly does the market form demand? The suspect requirement is convexity, without which a maker is said to be exploitable (Angeris and Chitra 2020). Sections 2–4 show the requirement is weaker and more informative than that.

Throughout, a scalar security settles at $\varphi(\omega) = \omega \in [-1, 1]$; vector statements substitute the convex hull of payoff vectors. Reference implementations and numerical theorem tests accompany the note in the mechanisms repository.

2 Coherence is a chord condition

Proposition 1 (no-arbitrage without convexity). *For a path-independent maker with cost C (no convexity assumed), there is no outcome-independent profitable fill from any state if and only if every chord slope of C lies in the convex hull of payoffs: for all q, s ,*

$$C(q + s) - C(q) \geq \min_{\omega} \varphi(\omega) s \quad (\text{scalar: } C \text{ is 1-Lipschitz}).$$

Proof. A fill s from state q has sure profit $\min_{\omega} \varphi(\omega) s - [C(q + s) - C(q)]$; positivity for some pair (q, s) is exactly the failure of the displayed inequality, and the inequality for all pairs says every chord slope is supported by the hull. ■

Two remarks. Cycling was never the exploit: the charge telescopes over any closed path whatever C is, so round trips are refunds; the exploit, when the chord condition fails, is the accumulation of sure-profit net positions at states where quotes exit the hull. And convexity enters the standard axiomatics not through no-arbitrage but through *information incorporation* (the marginal-cost monotonicity condition of Abernethy et al. (2013)); dropping that axiom while keeping the chord condition leaves a coherent non-convex maker, exhibited numerically in the companion repository. Arbitrage theory without convexity is developed in a different formalism by Lépinette and Tran (2017).

3 The market trades the biconjugate

Let $\hat{C}$ denote the lower convex envelope of a chord-coherent cost C (its biconjugate on the line (Rockafellar 1970)) and $g = C - \hat{C} \geq 0$ the gap.

Proposition 2 (contact and pass-through). *A myopic risk-neutral trader with believed mean μ in the hull, facing the maker at state q : (i) has optimal fills landing on the contact set $\{C = \hat{C}\}$, and (ii) earns maximal expected profit*

$$\Pi_C(q, \mu) = \Pi_{\hat{C}}(q, \mu) + g(q),$$

the envelope profit plus the gap at the starting state.

Proof. $\sup_s \mu s - [C(q + s) - C(q)] = \sup_x [\mu x - C(x)] - \mu q + C(q)$. Since $C \geq \hat{C}$ with equality on the contact set, and a maximizer of an affine function minus C is a point where an affine minorant touches C , hence touches $\hat{C}$, the supremum equals $\sup_x [\mu x - \hat{C}(x)]$ and is attained on the contact set. Adding and subtracting $\hat{C}(q)$ gives (ii). ■

The maker behaves observationally like its convex envelope. Concave stretches are unquotable intermediate states, holes in the book that rational flow jumps across; whoever lands inside one (noise) overpays the gap at the landing state, and by (ii) the next rational trader recoups it, so the maker is a conduit keeping envelope differences over any rational-to-rational span, and off-contact states are transient. The trade-set analogue on the CFMM side is the canonical concave trading function: an arbitrary invariant is behaviorally equivalent to a concave one (Angeris et al. 2024), with the limits of concavification mapped by Frongillo et al. (2024). Read as beliefs, a hole is a multimodal quote: the maker takes prices on either side of the gap and refuses every price inside, jumping discontinuously as flow accumulates, the book-side shape of the bimodal implied densities documented around binary events (Melick and Thomas 1997; Clark and Amen 2017).

4 Frictions price the remaining failure

The failure mode surviving §2 is chord slopes exiting the hull.

Proposition 3 (fees buy bounded incoherence). *If chord slopes exit the hull by at most ε per unit, i.e. $C(q + s) - C(q) \geq \min_{\omega} \varphi(\omega)s - \varepsilon|s|$, then a proportional fee $f \geq \varepsilon$ restores no-arbitrage.*

Proof. Sure profit with the fee is $\min_{\omega} \varphi(\omega)s - \Delta C - f|s| \leq (\varepsilon - f)|s| \leq 0$. ■

This is the mechanism-level counterpart of a theorem of mathematical finance: arbitrarily small proportional transaction costs restore no-arbitrage for price processes that are arbitrageable frictionlessly, with a consistent price system inside the spread as certificate (Guasoni 2006; Guasoni et al. 2010). In prediction markets the precedent is fees sized to expected arbitrage profit restoring bounded loss under privacy noise (Cummings et al. 2016; Frongillo and Waggoner 2018). The minimum viable spread of a cost is its arbitrage depth, and a venue quoting a wide spread to cover a wild landscape is coherent but uninformative in proportion: the market prices the model’s incoherence as uncertainty.

Lemma 4 (the fee lemma, absorbed). *For any chord-coherent C with envelope $\hat{C}$, the fee-bearing maker’s no-trade band at state q is the interval of beliefs within f of the envelope’s marginal price, and it is non-empty exactly on the contact set; at off-contact states every belief in the hull yields profit at least $g(q)$.*

Proof. By Proposition 2, profit at belief μ is the envelope profit plus $g(q)$, and with the fee the envelope profit is the soft-thresholded conjugate of the convex case, zero on the band around $\hat{C}'(q)$; on the contact set $g(q) = 0$ and the band survives, off it the additive $g(q) > 0$ leaves no zero. ■

The convex case, contact set everywhere, is the fee–spread lemma of the companion note *Combining Linear-Fee Market Makers*; conjugation reads only biconjugates, so the lemma survives non-convexity verbatim with the envelope in place of the cost.

Proposition 5 (a deep co-quoter is Moreau smoothing). *Merging the C -maker with a quadratic co-quoter of liquidity λ yields the venue with cost the Moreau envelope $e_{\lambda}C(x) = \min_y C(y) + (x - y)^2/(2\lambda)$, whose arbitrage depth decreases to zero as $\lambda \rightarrow \infty$; generically (tilted gaps) the merged venue is exactly convex at finite depth, while a symmetric double well retains a concave kink of magnitude $O(1/\lambda)$ at every depth, priced by a fee of the same order.*

Proof sketch. The merge is infimal convolution (Bhaskara et al. 2023), and inf-convolution with the quadratic is the Moreau envelope (Rockafellar 1970). $e_{\lambda}C$ is a minimum of convex branches $y \mapsto C(y) + (\cdot - y)^2/(2\lambda)$; non-convexity survives only at branch crossings, where

the slope jump is the branch-minimizer separation over λ , hence $O(1/\lambda)$, vanishing or merging entirely when the crossing is eliminated by tilt. ■

The two repairs are the two market primitives again: friction (ℓ_1 , the fee) and participant depth (ℓ_2 , capital), lasso and ridge. Optimization already pays them: a proximal step charges $\|\Delta\theta\|^2/(2\eta)$ per move, a trust region is an infinite fee outside a band, weight decay is a zero-quoting participant. Path-dependent optimizers correspond to makers whose quotes depend on flow history, the adaptive-liquidity territory where path independence is deliberately traded away: no maker can combine path independence, translation invariance and liquidity sensitivity (Othman et al. 2013), the adaptive class is axiomatized by Li and Wortman Vaughan (2013) with the homogeneous-risk-measure characterization in Othman and Sandholm (2011), no trade-history maker achieves every desideratum at once (Abernethy et al. 2014), and liquidity selection itself can be run as online learning (Nueve et al. 2026; Nueve and Waggoner 2025).

5 The characterization

A predictor is a market when its cost has finite arbitrage depth relative to affordable friction; convexity is the $f = 0$ special case. Within the coherent class, non-convexity spends expressiveness, the contact set in place of a full quoting range, and spends nothing else against rational flow. What the market form adds to a predictor is not new estimates but three things the estimation literature obtains otherwise: tuning constants become prices set by competition rather than formulas, composition becomes an algebra (merge as infimal convolution, chains as products), and the procedure becomes robust to strategic data, since every source pays to move the state. A market is the incentive closure of a predictor; a predictor is a market with the strategyproofing stripped out.

6 Open problems

Arbitrage depth as a capacity measure. The minimum viable spread of a trained landscape is computable; whether it tracks quantities learning theory already names (sharpness, mode connectivity) is open.

Two-agent pumps. Adversarial-training oscillation resembles a money pump between path-dependent agents, and gradient penalties resemble spread-widening; a precise statement is needed or the analogy retired.

Noise as subsidy. Minibatch noise lands states off-contact and the next informed step recoups the gap; whether SGD noise plays the economic role of noise traders awaits a model.

Books around binary events. The multimodal-quote reading of book holes (§3) is testable against limit-order data around court rulings and FDA decisions, where bimodal implied densities are already documented (Clark and Amen 2017).

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